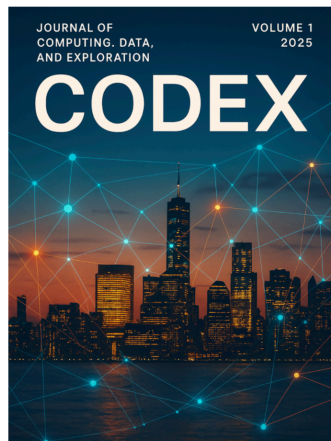


Student-Faculty Co-Creation of Open Educational Resources for Learning Applied Statistics with Open Source Software Tools

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Abstract

In most university courses, students learn from textbooks they did not help develop. We present an innovative approach where undergraduate students collaborated with faculty to create an open-access, interactive web-based e-book for an introductory applied statistics course. This process transforms undergraduates into co-authors rather than passive readers, fosters an appreciation for reproducible research, encourages academic collaboration, and offers diverse, hands-on opportunities to acquire and apply new skills and technologies. Faculty also benefit from fresh ideas and new perspectives. This type of collaboration is important to explore, as evidence on student-faculty partnerships in developing course materials is sparse in the literature, particularly in a Canadian context and in the area of statistics. In this paper, we illustrate the contributions of undergraduate students to curricular innovation in the development of the e-book by describing the development process, methodology and software tools used, and reflect on our experience as collaborators alongside faculty. Our participation deepened our learning, while producing resources to benefit future learners. We highlight the potential of open-source technologies and student-faculty collaboration to support the development of adaptive, interactive, and accessible resources for statistical education.



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